

WEEK 8

Anti-Forensics, Reporting & Core Analysis

Consolidated Findings Report using Autopsy

Tool	Autopsy - reporting and artefact review
Focus	Key Windows artefacts, anti-forensic indicators, interpretation and limitations
Evidence Type	Training Windows forensic case / previously collected user activity evidence
Method	Review -> Correlate -> Tag -> Generate Report -> Interpret
Scope Note	This Week 8 report covers the analysis task only; project-delivery administration is intentionally excluded.

Prepared as a practical student lab report

1. Purpose and Scope

Objective. The purpose of Week 8 was to consolidate the main forensic findings from the Windows case, review them for possible anti-forensic behaviour, and produce a clear report using the Autopsy report module. The emphasis was not on collecting new evidence, but on explaining what the already-extracted artefacts mean when they are considered together.

Approach. I reviewed the case results in Autopsy, checked the original artefact locations where possible, compared timestamps and related activity, tagged the most useful items, and generated a report from the selected results. Autopsy supports report generation from case results and can produce formats such as HTML and Excel; reports are then available under the Reports node in the case. [1]

Main idea. A single artefact was not treated as final proof. Instead, the conclusion was based on correlation. For example, a recent-document entry becomes stronger when it is supported by application execution traces, a Prefetch record, a logon session, or a removable-device artefact around the same time.

2. Autopsy Workflow Used

Step	Action	What I Did
1	Open Case	Opened the existing Windows case and confirmed the correct data source.
2	Review Ingest Results	Checked Recent Activity, file-system results, event logs, deleted files and other extracted artefacts.
3	Identify Key Evidence	Selected artefacts that directly supported user activity, program execution, USB use or deletion.
4	Check Anti-Forensic Indicators	Looked for cleared logs, suspicious deletion, inconsistent timestamps, gaps in activity and possible clean-up behaviour.
5	Correlate	Compared artefacts by user,

Step	Action	What I Did
		path and time instead of relying on one source only.
6	Tag Important Items	Tagged findings that should appear in the final case summary.
7	Generate Report	Used the Autopsy reporting function to export relevant case results for review. [1]

3. Consolidated Artefact Findings

The following artefacts were treated as the most useful for the consolidated findings section. The exact value in a real case must always be checked against the original data source and Autopsy result view.

Artefact	Location	What It Shows	Interpretation
RecentDocs	NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs	Recent file/folder references	Shows that a filename or document type was recently referenced by the user profile. Useful for identifying documents of interest, but it does not by itself prove the full file was opened or edited.
RunMRU	NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\RunMRU	Commands typed in Run dialog	Useful for showing user-entered commands or paths. Stronger when the command is also supported by execution artefacts or logs.
UserAssist	NTUSER.DAT\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist	GUI application execution traces	Supports that an application was launched through normal Windows shell interaction. ROT13-encoded value names should be decoded before interpretation.
Prefetch	C:\Windows\Prefetch*.pf	Program execution metadata	Can support execution of applications and provide run-related timing information. It is a strong supporting artefact when correlated with UserAssist or event activity.
USBSTOR	SYSTEM\CurrentControlSet\Enum\USBSTOR	USB storage device identity/history	Can identify removable storage that was known to the system, including vendor/product/serial information in many cases. Presence does not

Artefact	Location	What It Shows	Interpretation
			automatically prove data was copied.
Windows Event Logs	C:\Windows\System32\winevt\Logs\	Logon, process, system and audit events	Provides timestamped operating-system events. Event ID 1102 in the Security log indicates that the audit log was cleared and should be investigated as a possible anti-forensic action. [3]
Recycle Bin / Deleted Files	\$Recycle.Bin and file-system deleted-file views	Deleted names, paths and metadata	Deletion may be ordinary user activity or an anti-forensic attempt. Interpretation depends on timing, file importance and whether other evidence suggests deliberate concealment.
Timeline	Autopsy Timeline view	Cross-source chronological activity	Useful for comparing events before and after a suspicious action. Autopsy timeline combines timestamps from files, web artefacts and other extracted data. [2]

4. Example of Evidence Correlation

A finding becomes more reliable when several artefacts support the same activity. A simple example from a Windows user-activity investigation is shown below.

Source	Possible Observation	Meaning When Correlated
RecentDocs	A document name appears in the user profile	The user account recently referenced that document name.
UserAssist / Prefetch	The associated application has execution evidence	Supports that the application used to access the document was executed.
Event Log	A user session exists around the same period	Places the activity inside an authenticated session.
USBSTOR	A removable device is known to the machine	If timing also matches, removable-media use becomes a relevant investigative lead.

Interpretation: The combined artefacts can support a reasonable activity sequence, but they still do not prove the content of a file was copied to a USB device unless file-system or other evidence directly connects the file and the removable media.

5. Anti-Forensics Review

Anti-forensics refers to actions intended to hide, destroy, alter or make forensic evidence harder to interpret. In this week, the focus was on recognising indicators rather than assuming that every deletion or missing artefact was malicious.

Indicator	How I Checked It	Interpretation
Security log clearing	Check Security.evtx for Event ID 1102.	High-interest indicator. Microsoft documents Event 1102 as the event generated when the Security audit log is cleared. [3]
Deleted / wiped files	Review deleted files, Recycle Bin and unallocated-space results.	Deletion can be normal. It becomes more suspicious when important files disappear shortly after related activity or when secure-wiping utilities are present.
Timestamp manipulation	Compare file MAC times with timeline, execution artefacts and surrounding events.	A timestamp that conflicts with several independent artefacts may indicate manipulation; file timestamps alone should not be treated as unquestionable. [4]
History or MRU cleanup	Look for unusually empty MRU/browser/recent activity compared with otherwise active use.	Absence is only an indicator. Windows settings, cleanup utilities, application design or retention limits can also explain missing history.
Evidence gaps	Look for unexplained periods in logs and timeline.	A gap can suggest log loss or clean-up but may also be caused by shutdowns, retention limits or incomplete acquisition.

Week 8 assessment. The consolidated review should report anti-forensic indicators only when supported by evidence. If the case does not contain Event 1102, suspicious wiping activity, or clear timestamp contradictions, the correct conclusion is that no confirmed anti-forensic action was

identified from the reviewed artefacts. This is stronger and more defensible than inventing an anti-forensic finding simply because the week title includes anti-forensics.

6. Autopsy Report Module - Practical Procedure

- Finish or pause ingest only after the artefacts needed for the report have been processed.
- Review the Results and Data Sources trees and tag the items that directly support the findings.
- Open the reporting function and choose a suitable report type. Autopsy documentation lists report formats including HTML and Excel, and allows reports to be based on all results or tagged results. [1]
- For a concise case report, prefer Tagged Results so the report contains the items that were actually interpreted rather than thousands of unrelated artefacts.
- Select the correct data source and artefact types, generate the report, then open it from the Reports node and check that the expected evidence is present. [1]
- Keep the Autopsy-generated output as the technical attachment, while the written findings section explains why each artefact matters.

7. Consolidated Findings - Student Write-Up

The Windows evidence set contained several artefacts that were useful for reconstructing user activity. RecentDocs and RunMRU were useful for identifying files, commands and paths associated with the user profile. These artefacts were treated as activity indicators rather than complete proof because Windows MRU data can be cleared, overwritten or created by normal application behaviour.

Application-execution evidence was supported by UserAssist and Prefetch. When the same application appeared in more than one source, confidence in the execution finding increased. Prefetch was especially useful as a supporting artefact because it is stored outside the user hive, which means it can sometimes remain even if user-specific history has been modified or removed.

USB-related registry evidence was reviewed to identify removable storage previously known to the system. Device identification is important in a user-activity case because external media may explain how files entered or left the computer. However, a USBSTOR entry alone does not prove that a particular document was copied. File timestamps, link files, shell artefacts, volume information or other evidence would be needed to make that conclusion stronger.

Windows event logs provided timestamped system and account activity that could be compared with registry and file-system artefacts. As part of the anti-forensics review, I specifically checked for signs of audit-log clearing. Windows Security Event ID 1102 is a direct indicator that the Security audit log was cleared. [3] When this event is absent, it is not correct to claim that the log was cleared without additional evidence.

Deleted-file and Recycle Bin results were also reviewed. File deletion by itself was not considered anti-forensics because normal users regularly delete files. The context is more important: the file type, its relation to the case, the deletion time, and whether the deletion occurred shortly after

access or execution. The final interpretation therefore relied on multiple artefacts instead of a single deleted entry.

Overall, the strongest outcome of the week was the consolidated view. Autopsy made it easier to bring file-system, extracted artefact and timeline information into one case and then report selected results. The report module is useful for exporting case information, but the investigator still has to explain the meaning and limitations of the evidence. [1][2]

8. Limitations

- The analysis is limited to the evidence that was acquired. Volatile RAM data and network traffic are not reconstructed from a disk image unless they were collected separately.
- Some Windows artefacts have retention limits and may be overwritten during normal system use.
- Timestamps may be affected by time-zone settings, clock changes, copying behaviour or deliberate manipulation. Timeline evidence should therefore be correlated with other artefacts.
- The absence of an artefact is not proof that an activity never occurred. Logging configuration, cleanup tools, application settings and incomplete acquisition can all create gaps.
- Autopsy parsing depends on supported file formats and ingest modules. Important raw files should be checked manually when an automated result looks incomplete or inconsistent.
- Deleted-file recovery may be partial, especially after data has been overwritten or storage-device trimming has occurred.

9. Conclusion

Week 8 was mainly about turning technical artefacts into a defensible case explanation. The important lesson was that reporting is not the same as simply exporting data. Autopsy can collect and report large amounts of information, but the final findings must explain the source, location, meaning and weakness of each artefact. The strongest findings are those supported by more than one independent source. Anti-forensic behaviour should be reported carefully and only when there is enough evidence to support it.

10. Presentation Talking Points

- Start with the investigation question: what user activity can be supported by the artefacts?
- Show 5-7 key artefacts rather than every result in Autopsy.
- For each artefact, explain: location -> finding -> meaning -> limitation.
- Show the anti-forensics checks: log clearing, deleted files, timestamp anomalies and history gaps.
- Explain why correlation is more reliable than one artefact by itself.
- Finish with the main conclusion and one or two limitations.

References

- [1] The Sleuth Kit, "Autopsy User Documentation 4.23.0 - Reporting," Autopsy User Guide, accessed Aug. 2026.
- [2] The Sleuth Kit, "Autopsy Timeline Analysis," Autopsy documentation, accessed Aug. 2026.
- [3] Microsoft, "1102(S): The audit log was cleared," Microsoft Learn, Windows Security Auditing documentation.
- [4] The Sleuth Kit, "Autopsy Timeline Analysis Help," note on file timestamps and the possibility of timestamp modification.